

Futurism (Article) — ChatGPT is Consuming a Staggering Amount of Water.

Posted by u/BrimstoneDiogenes • 0 points • 16 comments

“Researchers from the University of Colorado Riverside and the University of Texas Arlington have shared a yet-to-be-peer-reviewed paper titled ‘Making AI Less Thirsty’ that looks into the environmental impact of AI training, which not only needs copious of electricity but also tons of water to cool the data centers.”

Comments

u/FreakLipsHighC • 1 points • 2023-04-14 21:21 UTC

It seems to me that concerns about water in California are pretty overblown. The ocean is right there. Desalination is going to be necessary, and it will cost energy but I think ultimately we're going to need to significantly increase our energy efficiency regardless, and that will need to be part of the deal

Now if I'm in Vegas I'm concerned

u/watevauwant • 5 points • 2023-04-14 12:01 UTC

How does it “consume” the water? Why is the water wasted? Does the water become toxic through cooling process? I doubt that. I don't get it... this article explains almost nothing

u/worstusername_ever69 • 8 points • 2023-04-14 11:45 UTC

Meanwhile some redditor: Now it needs water too so it's sentient

u/wavefield • 1 points • 2023-04-14 11:25 UTC

How is this not solvable? Build a giant air heat exchanger

u/animefreak701139 • 5 points • 2023-04-14 10:59 UTC

I'm pretty sure that water goes back into circulation so I don't think it's as big of an issue as that headline is making it out to be and full disclosure I did not actually read the article so if they went into that my apologies

u/Feo_daron • 1 points • 2023-04-14 11:22 UTC

The article does address your issue but here are my two cents. Generally, you're right in that the water used eventually goes back into the hydrological cycle, though that does not negate the necessity for addressing the water footprint of ChatGPT (and other large AI models) because:

a) Direct reason: water is a finite resource and its availability and quality vary depending on the region and time. If you keep withdrawing large amounts of water from one region and

transporting it to another, eventually the former region will experience water scarcity which will more likely impact the water supply of the local communities.

b) Indirect reason: the water in question needs to be extracted, treated, transported, etc which add to the overall energy consumption of the data centers which exacerbate their CO2 emissions expediting climate change.

u/genbetweenner • 1 points • 2023-04-14 11:55 UTC

Where does it address the question? I can't find where it talks about the fact that the water is reused (or not.)

u/Feo_daron • 1 points • 2023-04-14 12:24 UTC

The question is addressed in the article itself. Here's just a glimpse of one of the introductory paragraphs:

‘As illustrated in Figure 3, there are two water loops: one closed loop between the chiller and data center server room, and one open loop between the cooling tower and the chiller. Within the closed loop, water is not lost — it is pumped from the chiller into the data center to cool down the air handling unit’s supply air in order to maintain a proper server inlet temperature, and warm water that absorbs the server heat returns to the chiller direction. Through a heat exchanger at the chiller, the heat is transferred from the closed loop to the open loop. Note that the chiller may operate in a “bypass” mode for energy saving when the outside temperature is low. Along the open loop, some of the water gets evaporated (i.e., “consumed”) in the cooling tower to dissipate heat into the environment. Additionally, there is a process called “blown down” that drains the cooling water to reduce salt concentration accumulated in the cooling tower and hence also consumes water: the higher water quality, the more cycles of concentrations (i.e., water recirculates more times before “blown down”) and hence the less blown-down water [36].’

FYI: I think people misinterpreted the meaning of the article as "AI models 'destroy' water". Obviously that's not the case, rather it tries to highlight that large AI models need to cut back on the consumption of water through energy efficiency means (e.g., running at night when the temperature is naturally lower).

u/genbetweenner • 1 points • 2023-04-14 14:12 UTC

Hmm, I don't think the figures are showing up in my mobile browser. I don't see any of that.

u/Admirable-Shift-632 • 1 points • 2023-04-14 11:44 UTC

But what prevents them from recycling the clean fresh water? It's not like they evaporate it or dump it, the amount of treatment needed isn't like what goes on after it's been consumed by people or animals

u/HansJoachimAa • 4 points • 2023-04-14 11:06 UTC

Fresh water is an issue many places. But yeah, I don't understand why they can't reuse the water. Maybe they need a cooling tower like nuclear plants to reuse it.

u/animefreak701139 • 0 points • 2023-04-14 11:11 UTC

So I'm not a big tech guy but would there be an issue using salt water cuz I understand freshwater is a scarce resource right now but there's plenty of salt water

u/HansJoachimAa • 2 points • 2023-04-14 11:14 UTC

The salt damages the pipes and clogs the system. You could boil it first, but that costs a lot of energy.

u/animefreak701139 • 1 points • 2023-04-14 11:18 UTC

I want to be really obtuse and say something like well make more durable pipes or pipes less prone to clogging but I feel like if the answer was that easy they would already be doing it

u/HalcyonAlps • 2 points • 2023-04-14 12:03 UTC

It's doable. It's just more expensive.

u/AutoModerator • 1 points • 2023-04-14 10:49 UTC

We kindly ask /u/BrimstoneDiogenes to respond to this comment with the prompt they used to generate the output in this post. This will allow others to try it out and prevent repeated questions about the prompt.

^(Ignore this comment if your post doesn't have a prompt.)

While you're here, we have a [public discord server](<https://discord.gg/NuefU36EC2>). We have a free Chatgpt bot, Open Assistant bot (Open-source model), AI image generator bot, Perplexity AI bot, GPT-4 bot ([Now with Visual capabilities!](<https://cdn.discordapp.com/attachments/812770754025488386/1095397431404920902/image0.jpg>))

####[So why not join us?](<https://discord.gg/r-chatgpt-1050422060352024636>)

PSA: For any Chatgpt-related issues email support@openai.com.

####[ChatGPT Plus Giveaway]

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| [Prompt engineering hackathon](<https://www.flowgpt.com/hackathon>)

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